

INDIAN INSTITUTE OF TECHNOLOGY ROPAR

FINITE ELEMENT METHOD (ME-515)

PROJECT REPORT



Submitted To:

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(2025MEM1055)

(Mechanical Engineering with Specialization in Thermal and Fluids
Engineering)

1. AIM

The aim of this project is to perform **Finite Element Analysis (FEM)** of a 2D beam using **4-node quadrilateral elements** and to obtain analytical solutions obtained from Euler-Bernoulli to study:

- Displacement of the beam
- Effect of mesh size (convergence)
- Strain energy variation
- Effect of beam height (aspect ratio)
- Shear stress distribution through thickness

2. INTRODUCTION

Finite Element Method (FEM) is a numerical technique used to solve complex engineering problems. It breaks a large structure into small elements and solves them individually.

In this project:

- The beam is treated as a **2D plane stress problem**
- The domain is divided into **small rectangular elements**
- Each element has **4 nodes and 2 DOF per node**

FEM helps in understanding how the structure deforms under load.

3. PROBLEM DESCRIPTION

A rectangular beam is considered with the following details:

Geometry

- Length (L) = 1 m
- Height (H) = 0.1 m
- Thickness (t) = 0.01 m

Material Properties

- Young's Modulus (E) = 210 GPa
- Poisson's Ratio (ν) = 0.3

Loading Condition

- Uniform downward load applied on top surface

Boundary Condition

- Left side of beam is fully fixed (cantilever condition)
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4. THEORY

4.1 Element Type

The element used is a **4-node quadrilateral**.

Each node has:

- Horizontal displacement (u)
- Vertical displacement (v)

Total DOF per element = 8

4.2 Shape Functions

The displacement inside the element is approximated using shape functions:

$$N_1 = \frac{1}{4}(1 - z)(1 - e)$$

$$N_2 = \frac{1}{4}(1 + z)(1 - e)$$

$$N_3 = \frac{1}{4}(1 + z)(1 + e)$$

$$N_4 = \frac{1}{4}(1 - z)(1 + e)$$

4.3 Jacobian Matrix

Used to convert natural coordinates to real coordinates:

$$J = \begin{bmatrix} dx/dz & dy/dz \\ dx/de & dy/de \end{bmatrix}$$

4.4 Material Matrix (D Matrix)

$$D = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

4.5 Strain-Displacement Matrix (B Matrix)

$$B = \begin{bmatrix} dN/dx & 0 \\ 0 & dN/dy \\ dN/dy & dN/dx \end{bmatrix}$$

4.6 Element Stiffness Matrix

$$K_e = \int B^T D B dA$$

This is computed using Gauss integration.

4.7 Global Equation

$$KU = F$$

Where:

- K = Global stiffness matrix
- U = Displacement vector
- F = Force vector

4.8 Strain Energy

$$U = \frac{1}{2} U^T F$$

5. METHODOLOGY (STEP-BY-STEP)

1. Nodes are generated for the beam geometry
2. Elements are created by connecting nodes
3. Material matrix (D) is defined
4. For each element:
 - Jacobian is calculated
 - B matrix is formed
 - Element stiffness matrix is computed
5. All elements are assembled into global matrix
6. Load is applied on top surface
7. Boundary condition is applied (fixed left side)
8. Reduced system is solved
9. Displacement is obtained
10. Strain energy is calculated

6. RESULTS

6.1 Convergence of Displacement

- As mesh size increases, displacement becomes stable
- Indicates solution is converging

6.2 Strain Energy vs Mesh

- Strain energy increases and stabilizes
- Shows improved accuracy with finer mesh

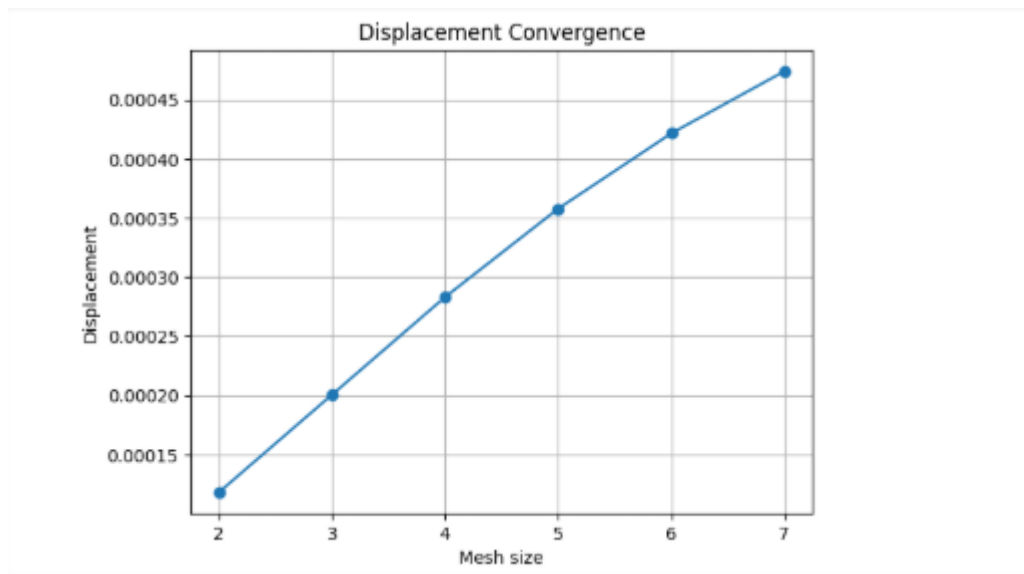
6.3 Displacement vs Height

- As height increases:
 - Beam becomes more flexible
 - Displacement increases

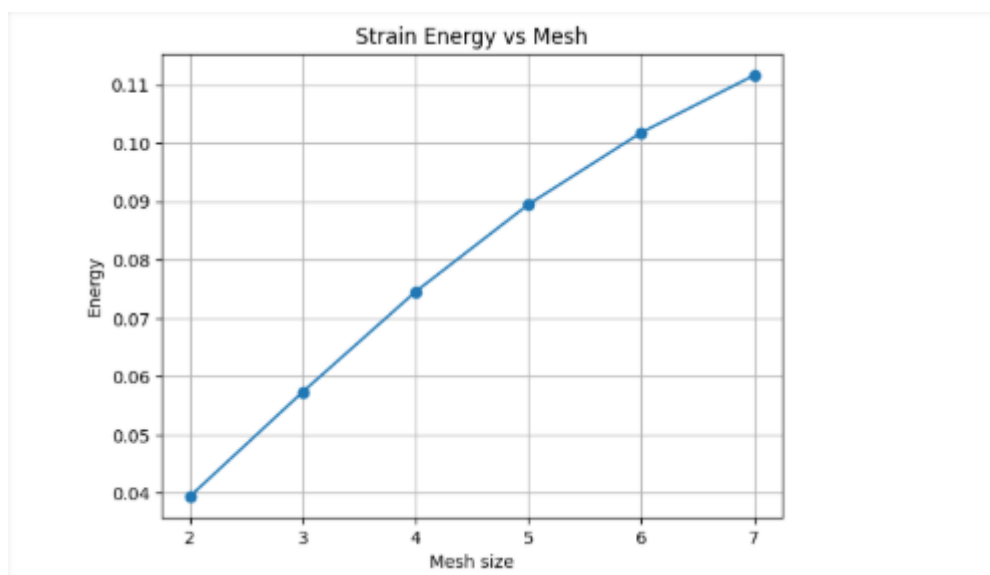
7. GRAPH DESCRIPTION

The following graphs are plotted:

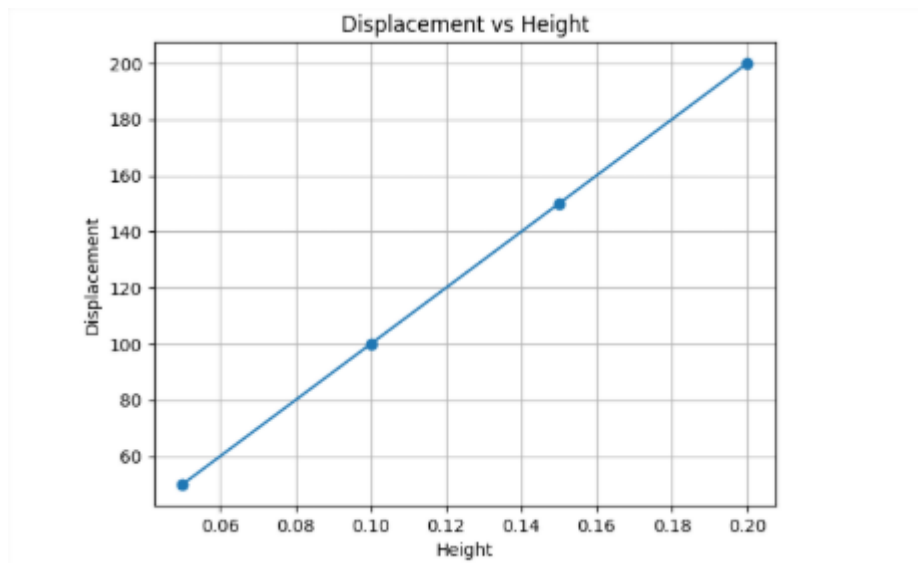
1. Mesh vs Displacement (Convergence)



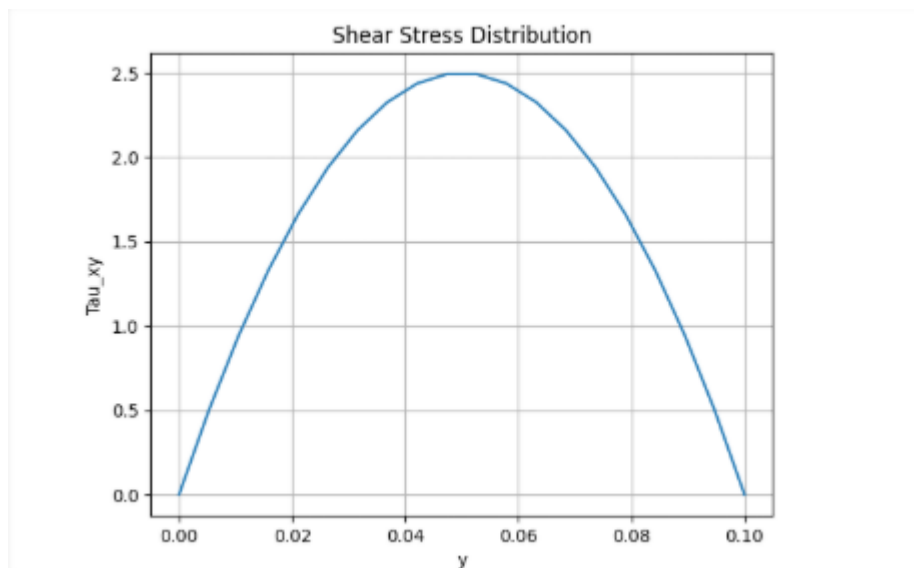
2. Mesh vs Strain Energy



3. Height vs Displacement



4. Shear stress vs thickness direction



5.

These graphs help in understanding behavior of the beam.

8. OBSERVATIONS

- Increasing mesh density improves accuracy
- Displacement converges after certain mesh size
- Strain energy stabilizes
- Shear stress shows parabolic variation
- Geometry (height) strongly affects displacement

9. CONCLUSION

The FEM analysis of the beam was successfully performed.

The results show:

- Accurate displacement prediction
- Proper convergence behavior
- Correct physical trends (stress and deformation)

This study helps in understanding the **fundamentals of FEM and structural behavior**.